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Experiment-8

EE1200 Electric Circuits Lab

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1 Objective

Design and implement a digital up/down counter that displays the number of people currently in the mess during peak lunch hours. The system will help students decide whether they can enter the mess based on its current occupancy, with the counter counting from 00 to 99.

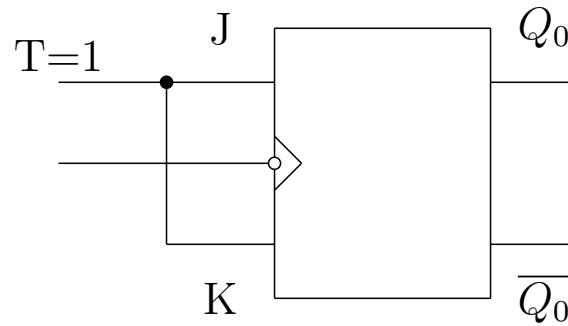
2 Components Required

- **T Flip-Flops:** 8 units (using JK flip-flops such as IC 7476, with J and K tied together as T inputs).
- **Logic Gates:** AND (IC 7411), OR (IC 7432), NOT (IC 7404).
- **Input/Output:** Push buttons for UP/DOWN control, two 7-segment displays, and two BCD decoders (IC 7447).
- Resistors, Jumping wires, Arduino , Breadboard

3 Theory

3.1 T Flip-Flop from JK Flip-Flop

- Conversion equation: $J = K = T$.
- Characteristic equation: $Q_{n+1} = Q_n \oplus T$.
- $T = 1$: Toggle state; $T = 0$: Maintain state.



3.2 Mod-10 Counter Design

3.2.1 State Transition Logic

- 4 flip-flops are required for decade counting (0000 to 1001).
- UP sequence: $0 \rightarrow 1 \rightarrow 2 \rightarrow \dots \rightarrow 9 \rightarrow 0$.
- DOWN sequence: $9 \rightarrow 8 \rightarrow 7 \rightarrow \dots \rightarrow 0 \rightarrow 9$.

3.3 Detailed Derivation of T Input Logic

3.3.1 1. UP Counter ($0 \rightarrow 9$)

State Transition Table (UP Mode)

Decimal	Present State	Next State	T_3	T_2	T_1
0	0000	0001	0	0	0
1	0001	0010	0	0	1
2	0010	0011	0	0	0
3	0011	0100	0	1	1
4	0100	0101	0	0	0
5	0101	0110	0	0	1
6	0110	0111	0	0	0
7	0111	1000	1	1	1
8	1000	1001	0	0	0
9	1001	0000	1	0	0
Note: T_0 is always 1.			$T_0 = 1$		

Karnaugh Maps for UP Mode

T_0 :

		CD			
		00	01	11	10
AB	00	1	1	1	1
	01	1	1	1	1
	11	1	1	X	X
	10	X	X	X	X

$$T_0 = 1$$

T_1 :

		CD			
		00	01	11	10
AB	00	0	1	1	0
	01	0	1	1	0
	11	X	X	X	X
	10	0	0	X	X

$$T_1 = \overline{Q_3} \cdot Q_0$$

T_2 :

		CD			
		00	01	11	10
AB	00	0	0	1	0
	01	0	0	1	0
	11	X	X	X	X
	10	0	0	X	X

$$T_2 = Q_1 \cdot Q_0$$

T_3 :

		CD			
		00	01	11	10
AB	00	0	0	0	0
	01	0	0	1	0
	11	X	X	X	X
	10	0	1	X	X

$$T_3 = Q_2 \cdot Q_1 \cdot Q_0 + Q_3 \cdot Q_0.$$

Final UP Counter Equations

$$T_0 = 1$$

$$T_1 = \overline{Q_3} \cdot Q_0$$

$$T_2 = Q_1 \cdot Q_0$$

$$T_3 = Q_2 \cdot Q_1 \cdot Q_0 + Q_3 \cdot Q_0.$$

3.3.2 2. Down Counter ($9 \rightarrow 0$)

State Transition Table (DOWN Mode)

Decimal	Present State	Next State	T_3	T_2	T_1
9	1001	1000	0	0	0
8	1000	0111	1	1	1
7	0111	0110	0	0	0
6	0110	0101	0	0	1
5	0101	0100	0	0	0
4	0100	0011	0	1	1
3	0011	0010	0	0	0
2	0010	0001	0	0	1
1	0001	0000	0	0	0
0	0000	1001	1	0	0
Note: T_0 is always 1.			$T_0 = 1$		

Karnaugh Maps for DOWN Mode

		CD			
		00	01	11	10
AB	00	1	1	1	1
	01	1	1	1	1
	11	1	1	X	X
	10	X	X	X	X

T_0 : Constant 1.

		CD			
		00	01	11	10
AB	00	0	0	0	1
	01	1	0	0	1
	11	X	X	X	X
	10	1	0	X	X

T_1 :

$$T_1 = (Q_1 + Q_2 + Q_3) \cdot \overline{Q_0}$$

		CD			
		00	01	11	10
AB	00	0	0	0	0
	01	1	0	0	0
	11	X	X	X	X
	10	1	0	X	X

$$T_2 = \overline{Q_1} \cdot \overline{Q_0} \cdot (Q_3 + Q_2)$$

T_3 :

		CD			
		00	01	11	10
AB	00	1	0	0	0
	01	0	0	0	0
	11	X	X	X	X
	10	1	0	X	X

$$T_3 = \overline{Q_2} \cdot \overline{Q_1} \cdot \overline{Q_0}$$

Final DOWN Counter Equations

$$T_0 = 1$$

$$T_1 = (Q_1 + Q_2 + Q_3) \cdot \overline{Q_0}$$

$$T_2 = \overline{Q_1} \cdot \overline{Q_0} \cdot (Q_3 + Q_2)$$

$$T_3 = \overline{Q_2} \cdot \overline{Q_1} \cdot \overline{Q_0}$$

3.3.3 Summary of T Input Logic

UP Counter

$$T_0 = 1 \quad (\text{always toggle})$$

$$T_1 = \overline{Q_3} \cdot Q_0$$

$$T_2 = Q_1 \cdot Q_0$$

$$T_3 = Q_2 \cdot Q_1 \cdot Q_0 + Q_3 \cdot Q_0.$$

DOWN Counter

$$T_0 = 1 \quad (\text{always toggle})$$

$$T_1 = (Q_1 + Q_2 + Q_3) \cdot \overline{Q_0}$$

$$T_2 = \overline{Q_1} \cdot \overline{Q_0} \cdot (Q_3 + Q_2)$$

$$T_3 = \overline{Q_2} \cdot \overline{Q_1} \cdot \overline{Q_0}$$

3.4 Combining the Up and down counter

Direction Control

Use a direction pin D to select between up/down logic:

- $D = 1$: Up mode
- $D = 0$: Down mode

Combined T Inputs

$$\begin{aligned} T_0 &= 1 \\ T_1 &= (\overline{Q_3} \cdot Q_0 \cdot D) + ((Q_1 + Q_2 + Q_3) \cdot \overline{Q_0}) \cdot \overline{D} \\ T_2 &= (Q_1 \cdot Q_0 \cdot D) + (\overline{Q_1} \cdot \overline{Q_0} \cdot (Q_3 + Q_2) \cdot \overline{D}) \\ T_3 &= ((Q_2 \cdot Q_1 \cdot Q_0 + Q_3 \cdot Q_0) \cdot D) + (\overline{Q_2} \cdot \overline{Q_1} \cdot \overline{Q_0} \cdot \overline{D}) \end{aligned}$$

Implementation

Use AND-OR gates to merge up/down logic based on D .

3.5 Carry/Borrow Generation

Carry (Up Mode)

$$\text{CARRY} = (Q_3 \cdot Q_0) \cdot D$$

Borrow (Down Mode)

$$\text{BORROW} = (\overline{Q_3} \cdot \overline{Q_2} \cdot \overline{Q_1} \cdot \overline{Q_0}) \cdot \overline{D}$$

Tens Clock

$$\text{CLK}_{\text{tens}} = \text{CARRY} + \text{BORROW}$$

3.6 Tens Counter Design

- Identical to the units counter
- Clocked by CLK_{tens}
- Shares direction control pin D

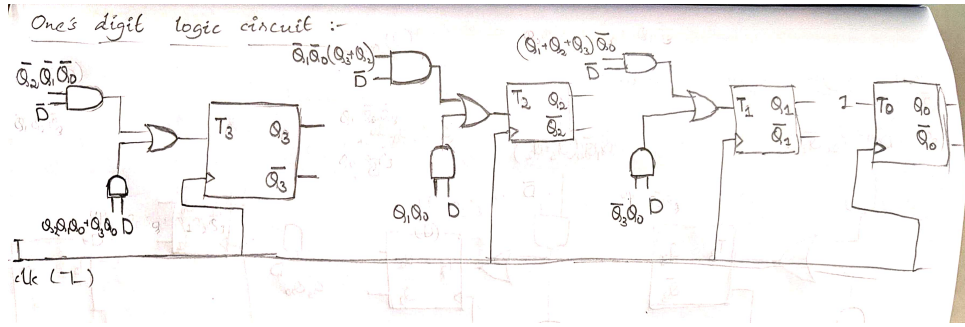


Figure 1: One's digit Logic circuit

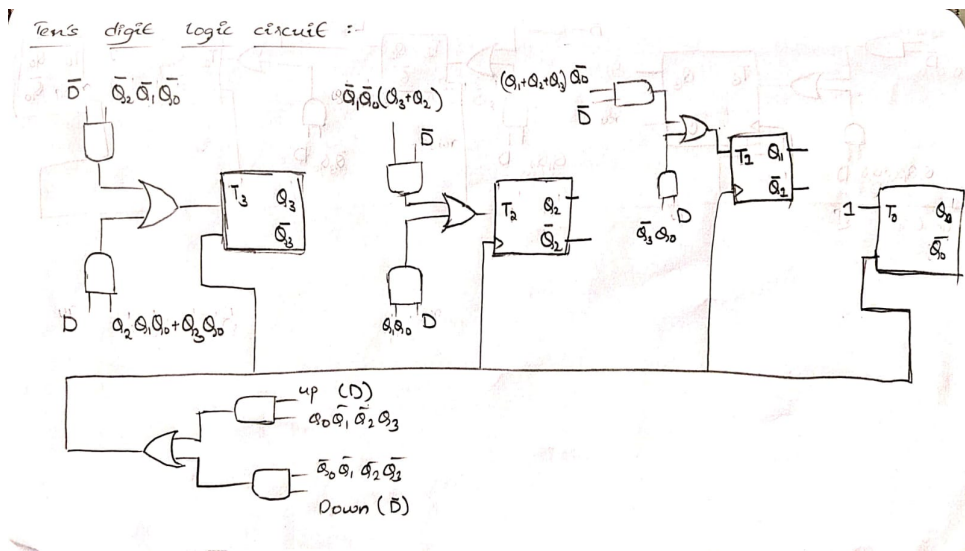


Figure 2: Ten's digit Logic circuit

3.7 Full Circuit Connections

- **Units Counter:** 4 T flip-flops with combined up/down logic, connected to a BCD decoder and 7-segment display
- **Tens Counter:** Duplicate of the units counter, clocked by CLK_{tens}
- **Direction Pin (D):** Connected to all AND-OR gates in both counters
- **Carry/Borrow Circuit:** Detects units = 9 (carry) or units = 0 (borrow) and generates CLK_{tens}

3.8 Full Capacity Logic

- An alert is activated when both counters display 9 (i.e. 1001 on each).

$$Full = (Q_{3u} Q_{0u}) \cdot (Q_{3t} Q_{0t})$$

- A visual indicator (e.g., LED) turns on, and an audio alert (buzzer) can be activated.

4 Procedure

1. Circuit Assembly

Step 1: Connect Buttons to Arduino

- **UP Button:** Connect one terminal to Arduino pin 2 and the other to GND. Enable INPUT_PULLUP in the code.
- **DOWN Button:** Connect one terminal to Arduino pin 3 and the other to GND.

Step 2: Connect Arduino Outputs to Counter Circuit

- **Direction Pin:** Connect Arduino pin 9 to the direction control input of the counter.
- **Clock Output:** Connect Arduino pin 8 to the clock input of the units counter.

Step 3: Counter Circuit Connections

- Configure 8 T flip-flops (IC 7476) for units and tens counters.
- Use AND/OR gates (IC 7408/7432) to implement the bidirectional T-input logic.
- Connect the carry (for up counting) or borrow (for down counting) signals from the units counter to the tens counter's clock input.
- Connect the output of each counter to a BCD decoder and 7-segment display.

2. Arduino Code Setup

Listing 1: Arduino Code

```
const int upButton = 2, downButton = 3, clockOut = 8, directionPin = 9;

void setup() {
    pinMode(upButton, INPUT_PULLUP);
    pinMode(downButton, INPUT_PULLUP);
    pinMode(clockOut, OUTPUT);
    pinMode(directionPin, OUTPUT);
    digitalWrite(clockOut, LOW);
}

void loop() {
    if (digitalRead(upButton) == LOW) {
        digitalWrite(directionPin, HIGH); // Up mode
        pulseClock();
    } else if (digitalRead(downButton) == LOW) {
        digitalWrite(directionPin, LOW); // Down mode
        pulseClock();
    }
}
```

```
}  
  
void pulseClock() {  
    digitalWrite(clockOut, HIGH);  
    delay(200); // Debounce delay  
    digitalWrite(clockOut, LOW);  
    while (digitalRead(upButton) == LOW || digitalRead(downButton) == LOW);  
}
```

Step 2: Verify Outputs

- Confirm that `directionPin` is HIGH when the UP button is pressed and LOW when the DOWN button is pressed.
- Check that `clockOut` generates a 200 ms HIGH pulse on button press.

3. Testing

Step 1: Test Units Counter

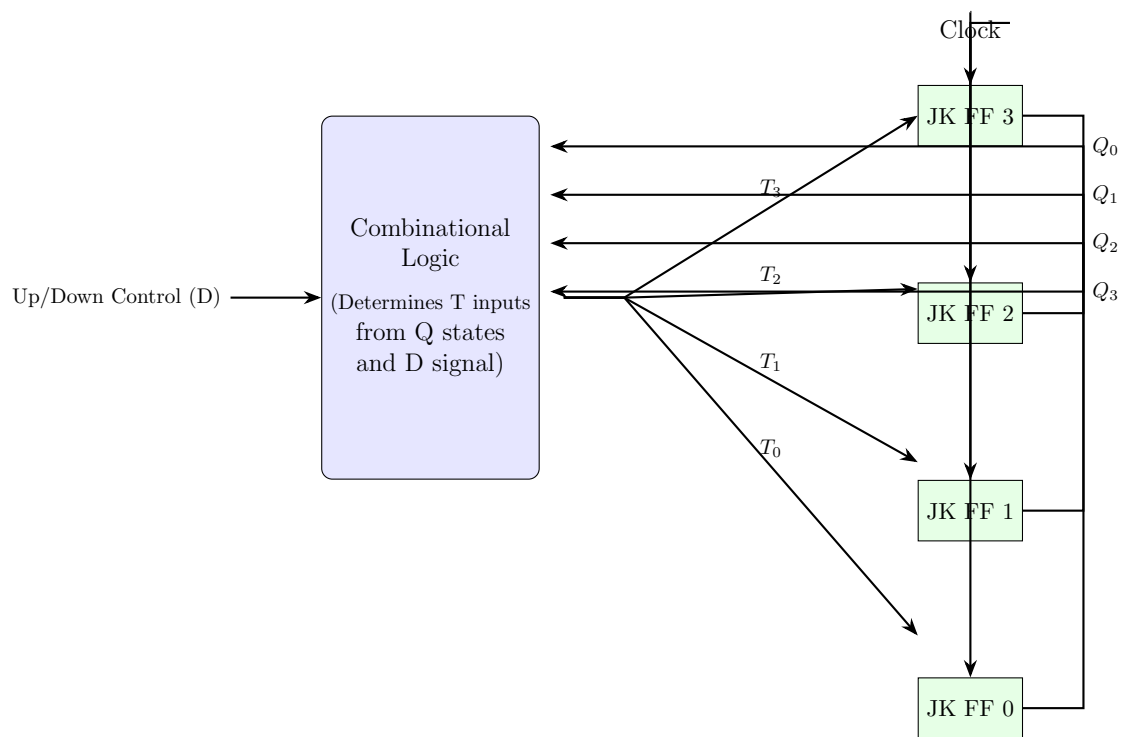
- Disconnect the tens counter.
- Press the UP button to verify the units counter increments from 0 to 9.
- Press the DOWN button to verify the units counter decrements from 9 to 0.

Step 2: Test Tens Counter

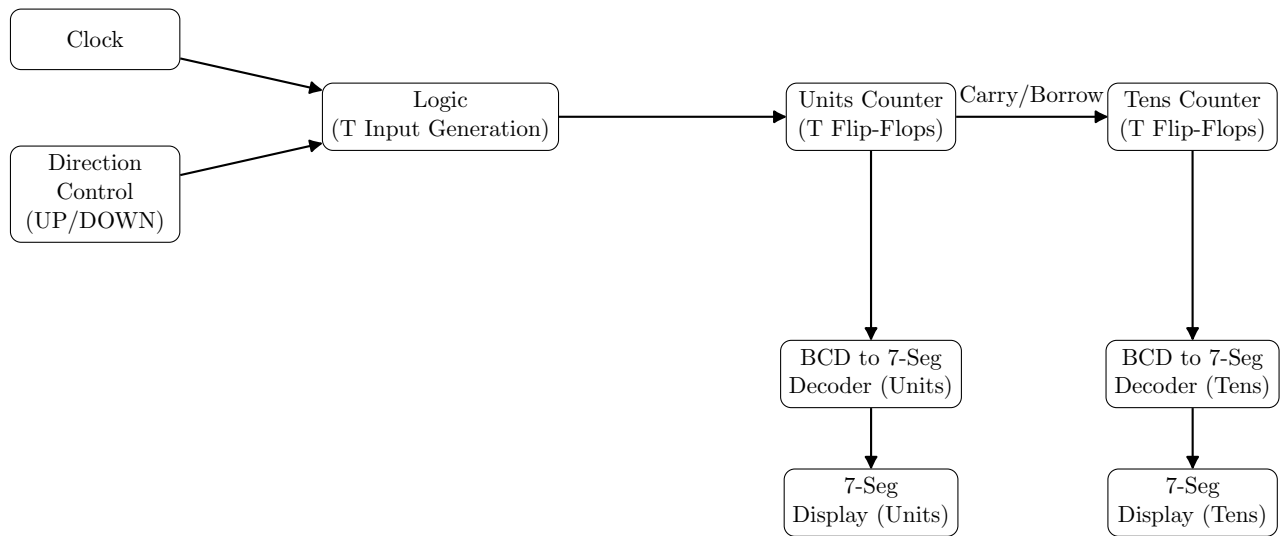
- Disconnect the units counter.
- Manually apply clock pulses to the tens counter to verify correct incrementation and decrementation.

Step 3: Full System Test (Mod-100)

- Reconnect units and tens counters.
- In Up mode, confirm that the sequence cycles from 00 to 99 and resets to 00 with the correct carry signal.
- In Down mode, confirm that the sequence cycles from 99 to 00 and wraps to 99 with the correct borrow signal.



All J & K inputs tied together (T configuration)



5 Circuit

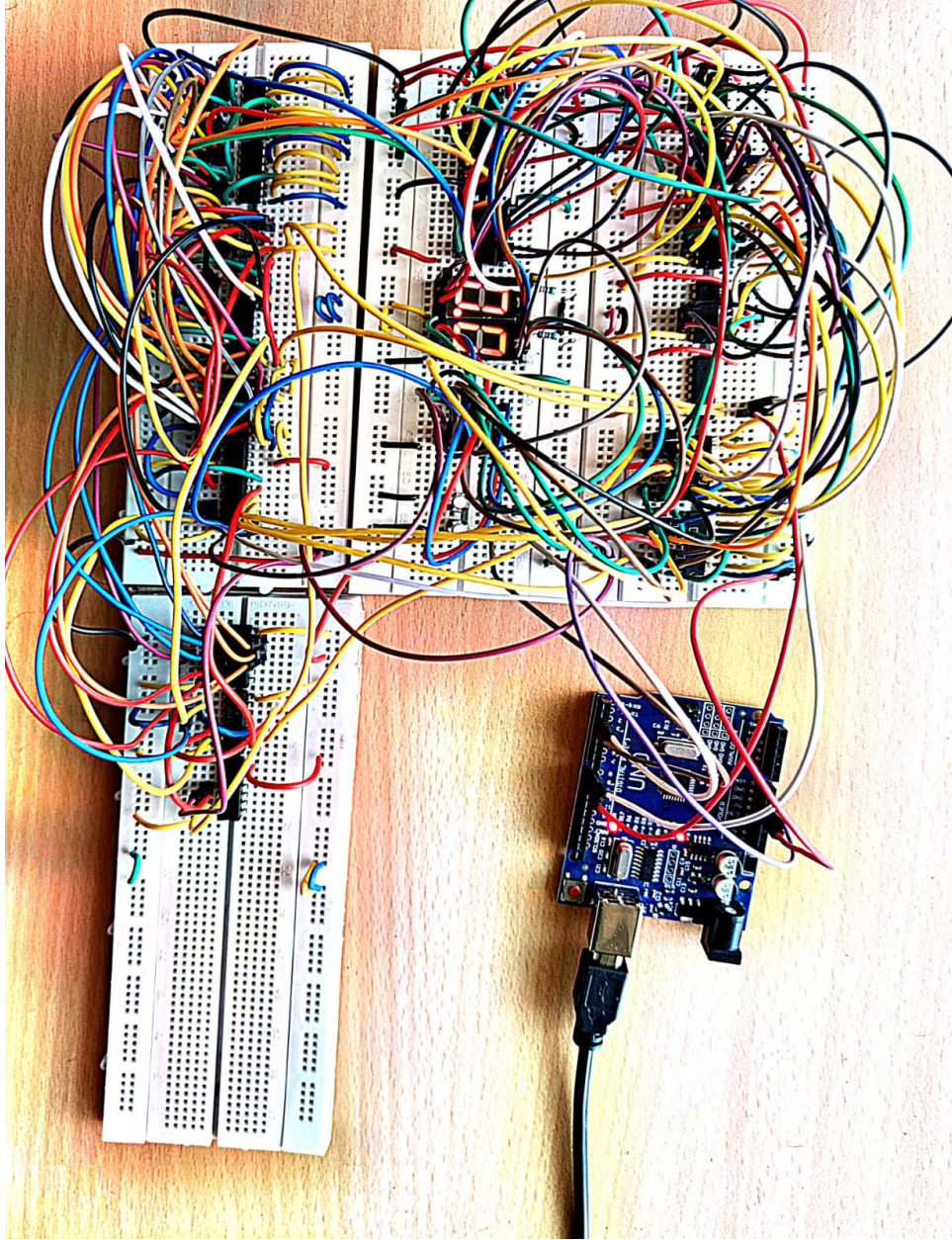


Figure 3: Circuit Diagram

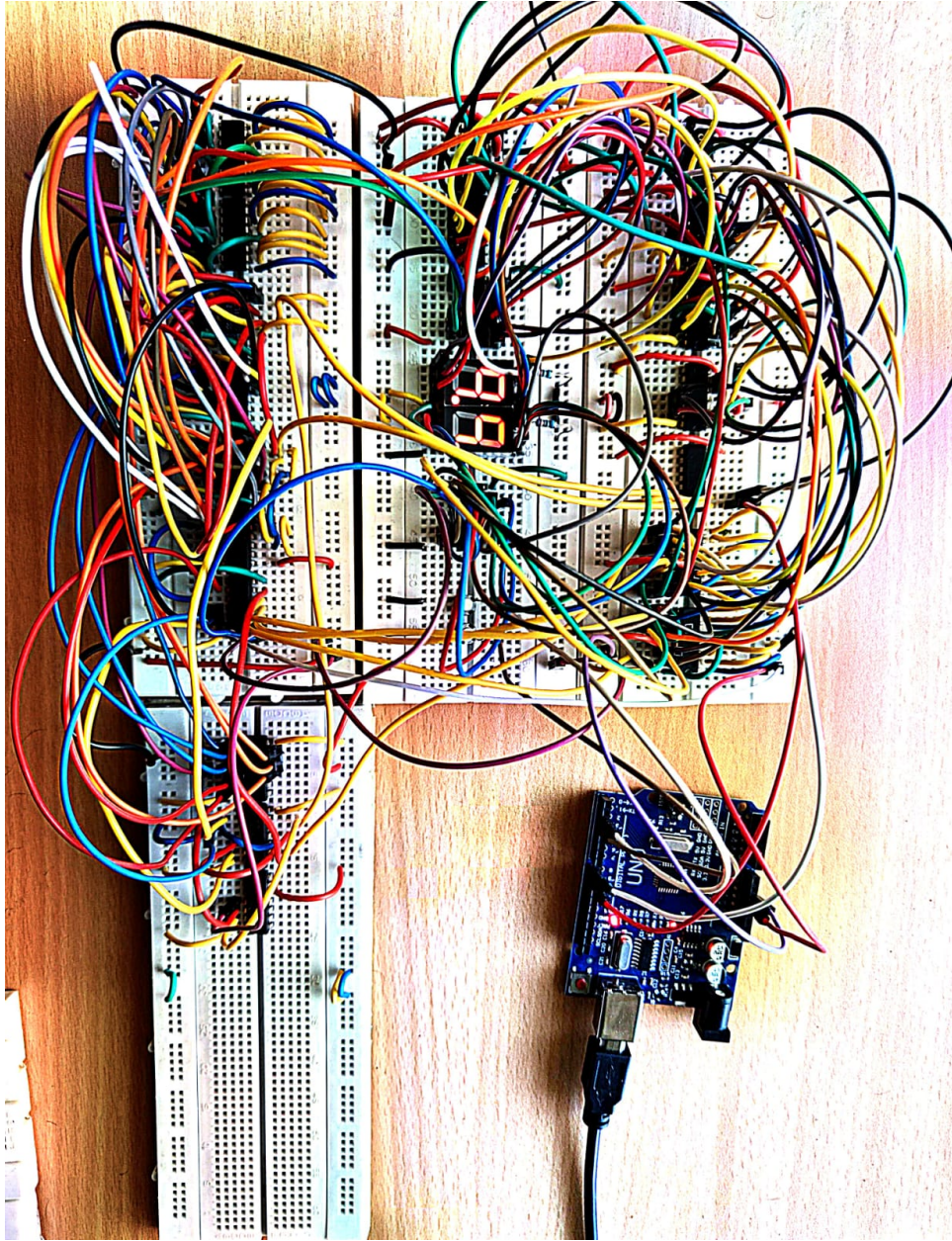


Figure 4: Circuit Diagram

6 Observations

- The counter used four JK flip-flops in T-mode.
- T inputs were derived from a combinational logic block using Q outputs and the control signal D .
- All flip-flops were clocked synchronously.
- When $D = 1$, the counter incremented; when $D = 0$, it decremented.
- Outputs Q_0 to Q_3 followed the expected binary sequence.

7 Result

- The 4-bit synchronous counter worked correctly in both up and down modes.
- Output matched theoretical binary sequences.
- The design validated the use of JK flip-flops for directional counting.

8 Conclusion

The digital up/down counter designed using T flip-flops has been successfully implemented for both mod-10 and mod-100 operations. Detailed derivations via state transition tables and Karnaugh map simplification provided optimized logic for the T inputs, ensuring glitch-free performance. This project demonstrates robust design principles for digital counters and effectively serves as an occupancy indicator for the mess during peak lunch hours.